

# Ratio, proportion and percentages - Set 2 - Solutions

Specification M3.1-M3.11.

## Answer key

| Question | Answer | Topic                             |
|----------|--------|-----------------------------------|
| 1        | D      | Ratio, proportion and percentages |
| 2        | F      | Ratio, proportion and percentages |
| 3        | E      | Ratio, proportion and percentages |

## Worked solutions

### Question 1 RAT-09 · Ratio, proportion and percentages · spec M3.9

$p$  is directly proportional to  $\sqrt{q}$ , and  $q$  is inversely proportional to  $r$ . Which of the following is true?

**Answer: D**

$$p = a\sqrt{q} \text{ and } q = \frac{b}{r}, \text{ so } p = a\sqrt{\frac{b}{r}} = \frac{a\sqrt{b}}{\sqrt{r}}.$$

### Question 2 RAT-11 · Ratio, proportion and percentages · spec M3.1

On a map with scale 1 : 25 000, a lake has area  $8 \text{ cm}^2$ . What is the actual area of the lake in  $\text{km}^2$ ?

**Answer: F**

1 cm represents 25 000 cm = 0.25 km, so  $1 \text{ cm}^2$  represents  $0.0625 \text{ km}^2$ .  $8 \times 0.0625 = 0.5 \text{ km}^2$ .

### Question 3 RAT-12 · Ratio, proportion and percentages · spec M3.6

In a year group the ratio of boys to girls is 3 : 4. Two boys leave and two girls join, and the ratio becomes 2 : 3. How many students are in the year group?

**Answer: E**

$$\text{With } 3k \text{ boys and } 4k \text{ girls: } \frac{3k-2}{4k+2} = \frac{2}{3} \Rightarrow 9k-6 = 8k+4 \Rightarrow k = 10. \text{ Total } 7k = 70.$$